

AI Interview Simulator

Comprehensive Project Architecture, Features & Technical Specification

Version: 2.0 | Stack: React + Vite + FastAPI + PostgreSQL + Google Gemini | Status: Production Ready

1. Executive Summary

The **AI Interview Simulator** is an enterprise-grade paired interview training and evaluation platform. It replicates end-to-end technical, behavioral, and DSA interviews under authentic corporate conditions. Featuring realistic speech-to-text conversational AI, video/microphone cockpit control, strict anti-cheating proctoring, and an interactive competitive coding arena populated with **390 authentic LeetCode and GeeksForGeeks problems**, the application bridges the gap between passive study and high-pressure live interview performance.

Core Pillars	Implementation Highlights
AI Interview Engine	Real-time voice recognition (Web Speech API) & natural speech synthesis (TTS) powered by Gemini AI with smart conversational pool fallback.
Coding Arena & Sandbox	390 problems mapped across 20+ algorithmic patterns, Monaco Editor, in-browser test runner, and automated test-case evaluation.
Proctoring & Integrity	Automatic fullscreen enforcement, tab-switch violation tracker (auto-submit after 3 warnings), and 90s inactivity countdown modal.
Deep Analytics & PDF	Automated candidate scorecards, mistake timeline, strengths/weaknesses breakdown, and instant one-click PDF report generation.

2. Technical Architecture & Technology Stack

Component Layer	Technologies & Libraries	Responsibilities
Frontend UI/UX	React 18, TypeScript, Vite, Tailwind CSS, Lucide React, Monaco Editor	Responsive modern cockpit, coding IDE, real-time webcam feed, voice visualizers, audio player.
Client Media & Voice	WebRTC MediaStream, Web Speech API (SpeechRecognition & SpeechSynthesis)	Continuous voice dictation, microphone level tracking, candidate camera stream, automatic device cleanup.
Backend Core	FastAPI, Python 3.12, Uvicorn, Pydantic v2, Python-JOSE (JWT)	High-concurrency async REST API, OAuth2 authentication, role-based access control, session lifecycle.
Database & ORM	PostgreSQL 16, SQLAlchemy 2.0, Psycopg2-Binary	Relational storage for candidates, interview sessions, answers, reports, 390 DSA questions, and submissions.
AI Intelligence	Google Gemini API (gemini-2.5-flash) + Smart Fallback Pool	Context-aware interview questions, dynamic follow-up generation, candidate response scoring & evaluation.
Execution Sandbox	Isolated Subprocess Runner (Python 3.12 venv & Node.js)	Safe code compilation, stdin parameter injection, stdout capturing, execution time & memory measurement.
Report Generation	ReportLab 4.1, PyPDF	Dynamic generation of branded, publication-quality PDF candidate scorecards with custom canvas pagination.

3. Core Functional Modules

A. Live Audio/Video Mock Interview Simulator

- **Pre-Interview Lobby:** Comprehensive camera and microphone testing with immediate feedback.
- **Dual Voice Interaction:** Candidate speaks naturally via speech-to-text; AI speaks questions aloud using natural TTS voice synthesis.
- **Conversational Adaptability:** Dynamically generates follow-up questions tailored to candidate's previous responses and resume.
- **Automatic Hardware Shutdown:** Automatically turns off webcam LED and terminates microphone recording the instant the session concludes.

B. Competitive Coding Arena (390 Curated DSA Problems)

- **Pattern-Based Learning:** Problems classified by 20+ algorithmic patterns (Sliding Window, Two-Pointer, Monotonic Stack, BFS/DFS, Heaps, DP, etc.).
- **Authentic Test Data:** Complete collection of 390 official LeetCode and GeeksForGeeks problems with genuine sample inputs, expected outputs, explanations, and constraints.
- **Intelligent Parameter Unpacking:** Starter templates parse standard inputs (e.g. `nums = [2,7,11,15]`, `target = 9`) and automatically inject arguments into the function call.
- **Dual Submission Handling:** Works seamlessly whether the candidate includes competitive programming ``main()`` blocks or submits function-only LeetCode solutions.

C. Anti-Cheating & Proctoring Engine

- **Browser Fullscreen Enforcement:** Auto-prompts fullscreen mode upon entering the live round.
- **Tab-Switch Detection:** Tracks document visibility events. Issues 2 visual warnings; 3rd violation terminates the interview immediately.
- **Inactivity Monitoring:** Detects user inactivity exceeding 90 seconds, triggering a 30-second countdown prompt before auto-submitting.

D. Detailed Evaluation & Mistake Timeline

- **Multi-Criteria Scorecard:** Overall Score (0-100), Technical Accuracy, Communication Clarity, Depth, and Problem Solving.
- **Question-by-Question Breakdown:** Candidate answers side-by-side with ideal answers, detected errors, and AI coaching notes.
- **Mistake Timeline:** Chronological log of pauses, weak explanations, keyword omissions, and proctoring events.

4. Database Architecture & Key Entities

Entity / Table	Key Attributes	Description & Relationship
users	id, email, hashed_password, full_name, role, created_at	Candidate and Admin credentials with bcrypt security and JWT auth.
interviews	id, user_id, target_role, difficulty, status, total_questions, score	Central interview session lifecycle: in_progress, completed, auto_submitted.
questions	id, interview_id, question_text, category, order_index, difficulty	Curated and Gemini-generated questions tailored to user role.
answers	id, question_id, candidate_answer, score, feedback, duration_seconds	Recorded speech transcription, latency, and individual question evaluations.
interview_reports	id, interview_id, overall_score, technical_score, strengths, weaknesses	Aggregated analytical report data with detailed feedback notes.
coding_problems	id, title, category, pattern, difficulty, examples, test_cases, templates	Catalog of 390 DSA problems with genuine inputs, outputs, and starter code.
coding_submissions	id, user_id, problem_id, code, language, status, runtime_ms, memory_kb	Candidate execution attempts, evaluation results (Accepted/WA), and runtimes.

5. Execution & Deployment Quick Start

1. Backend Setup & Startup:

```
cd backend
.\env\Scripts\activate
uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

2. Frontend Setup & Startup:

```
cd frontend
npm install
npm run dev (Available at http://localhost:5173)
```

3. Database Configuration:

```
PostgreSQL running at postgresql://postgres:postgres@localhost:5432/ai_interview_sim
```

6. Project Impact & Commercial Value

The AI Interview Simulator delivers a complete, professional training pipeline for candidates and an automated assessment tool for recruiters. Combining zero-latency speech recognition, Gemini-powered adaptive questioning, a battle-tested 390-problem coding platform, and anti-cheating proctoring, it is an end-to-end, market-ready AI SaaS product.